

**KOUSIK DAS**

**CURRICULUM VITAE**

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**Personal Profile :**

Name : **Dr. Kousik Das**

Current Designation : Institute Postdoctoral Fellow (IPDF)  
Department of Industrial Engineering & Operations Research,  
Indian Institute of Technology Bombay, India  
(Postdoc Supervisor- Prof. Veeraruna Kavitha)

Employee Code : 20004007

Duration : [August 2024 to Present](#)

Research Interests : Applied Probability, Queueing Theory (and Polling System),  
Stochastic Approximation Methods, Game Theory, Markov  
Decision Process, Multi-armed Bandit Problems, Reinforcement  
Learning

Gender : Male

Date of Birth : 24<sup>th</sup> December, 1993

Marital Status : Single

Permanent address : 38/B, Manujendra Dutta Road,  
Dum Dum Cantonment, 24 Parganas (North)  
Kolkata-700 028, West Bengal

Nationality : Indian

Caste : General

Giving Postdoc Interviews in [June and July 2024](#)

**Professional Experience:**

| Positions held      | Name of the Department and Institute                                         | From                        | To                         |
|---------------------|------------------------------------------------------------------------------|-----------------------------|----------------------------|
| Research visitor    | Neo Team, Inria Sophia Antipolis, France                                     | <a href="#">17/11/2025</a>  | <a href="#">21/11/2025</a> |
| Contractual faculty | Department of Mathematics, National Institute of<br>Technology Raipur, India | <a href="#">August 2023</a> | <a href="#">May 2024</a>   |

## Academic Qualification:

### Doctor of Philosophy

**Ph.D. Degree awarded on** [28/11/2023](#)

**Thesis submitted:** [30/06/2023](#)

**Joining date:** [21/08/2018](#)

**Supervisor:** Dr. Sujit Kumar Samanta

Associate Professor (Head of Department)

Department of Mathematics, National Institute of Technology Raipur

**Thesis title:** Computational aspects of discrete-time bulk queueing systems with correlation in arrival and/or service processes

**Course work marks:** 8.50 CGPA

### CSIR Sponsored Project Fellow

**Duration** : [May 2018 to August 2018](#)

**Designation** : Junior Research Fellow

**PI** : Dr. Sujit Kumar Samanta

Associate Professor (Head of Department)

Department of Mathematics, National Institute of Technology Raipur

**Project title** : Analytical and Computational Studies of Some Queueing Models with Markovian Arrival Process and /or Markovian Service Process

| Qualifying National Wide exam | Year of Passing            | Score   | AIR(MA) |
|-------------------------------|----------------------------|---------|---------|
| CSIR NET(JRF)                 | <a href="#">May 2018</a>   | 55.25%  | 84      |
| GATE                          | <a href="#">March 2018</a> | 32.67 % | 902     |

**Preparing for GATE, CSIR NET and PhD Entrance exams from** [October 2016 to April 2018](#)

| Degree                            | Year of Passing                | University/ Board | College/School                        | Marks (%) |
|-----------------------------------|--------------------------------|-------------------|---------------------------------------|-----------|
| M.Sc. (MATHS)                     | <a href="#">September 2016</a> | WBSU              | Dum Dum Motijheel College (P.G.-unit) | 83.3%     |
| B.Sc. (Hons)(MATHS)               | <a href="#">July 2014</a>      | WBSU              | Bidhannagar College                   | 61.38%    |
| Higher Secondary (10+2) (SCIENCE) | <a href="#">2011</a>           | WBCHSE            | Dum Dum Kishore Bharati High School   | 85.80%    |
| Secondary (10 <sup>th</sup> )     | <a href="#">2009</a>           | WBBSE             | Dum Dum Baidyanath Institution        | 88%       |

**Publication Details: SCIE/SCOPUS/ESCI Journals/A\* International Conference**

1. S. K. Samanta, K. Das, A simple procedure to determine the queue length and waiting time distributions for  $M/G^{a,b}/1$  queueing system, *Mathematical Methods of Operations Research*, 2025. (SCIE)
2. K. Avrachenkov, K. Das, V. Kavitha, V. Singh, Stability of polling systems for a large class of Markovian switching policies, *Proceedings of the 2025 64th IEEE Conference on Decision and Control*. (Accepted)
3. K. Das, V. Singh, K. Avrachenkov, V. Kavitha, Stability of polling systems: A linear class of two-phase switching policies near fluid limit, 2025. Abstract accepted in *34th European Conference on Operational Research (EURO 2025)*, University of Leeds, United Kingdom.
4. K. Das, S. K. Samanta, Analytical and computational investigation of the  $GI/D-BMSP/1$  queueing system, *Journal of Systems Science and Systems Engineering*, 2024. (SCIE)
5. K. Das, S. K. Samanta, Analytic computation for stationary distributions of  $D-BMAP/D-MSP^{(a,b)}/1$  queueing system, *International Journal of Computer Mathematics: Computer Systems Theory*, 2024. (SCOPUS/ESCI)
6. S. K. Samanta, K. Das, Detailed analytical and computational studies of  $D-BMAP/D-BMSP/1$  queueing system, *Methodology and Computing in Applied Probability*, 25(12), 2023. (SCIE)
7. K. Das, S. K. Samanta, Computational analysis of  $GI^{[X]}/D-MSP^{(a,b)}/1$  queueing system via  $RG$ -factorization, *Mathematical Methods of Operations Research*, 98(1):1-39, 2023. (SCIE)
8. S. K. Samanta, K. Das, Computing stationary distributions of the  $D-MAP/D-MSP^{(a,b)}/1$  queueing system, *Journal of Ambient Intelligence and Humanized Computing*, 13(1): 571-590, 2022. (SCIE)
9. K. Das, S. K. Samanta, Modelling and analysis of  $D-BMAP/D-MSP/1$  queue using  $RG$ -factorization, *Quality Technology & Quantitative Management*, 18(3): 355-381, 2021. (SCIE)

**Submitted Articles:**

1. R. Vyas, K. Das, V. Kavitha, S. Roy, Multi-type random game dynamics: limits at discontinuities and cyclic limits, 2025. (Submitted)
2. S. K. Samanta, A. K. Jena, K. Das, R. Nandi, Analysis of  $N$ -policy  $GI^{[X]}/MSP/1$  queueing system using  $RG$ -factorization, 2024. (Submitted)

**Reviewer of International Journals/ Conferences:**

- The Journal of Supercomputing
- International Journal of Data Science and Analytics
- International Journal of Applied and Computational Mathematics
- Scientific Reports
- CDC Conference in The IEEE Control Systems Society Conference, Journals, Award Management System

**Attended Conference/Workshop:**

- Participated in the “Lectures on Probability and Stochastic Processes XVII” in 2025 at the Department of Mathematics, Indian Institute of Technology Madras
- Participated and presented in the *34th European Conference on Operational Research (EURO 2025)*, University of Leeds, United Kingdom
- Participated in the “Reinforcement Learning workshop 2025” -IISc Bengaluru

- Participated in the *FDP on “Amplifying AI: Advanced Optimization Strategies for Enhanced Performance”*, 2024 -- Department of Computational Science & Humanities, IIIT Kottayam
- Participated and presented in the *International Conference on “Optimization, Learning and Analytics in Business (OLAB 2022)”* -- Department of Mathematics, Heritage Institute of Technology and Operational Research Society of India. Title of the paper: **Cost optimization of the N-policy  $GF^X/D-MSP/1$  queueing system via computational analysis of the queue**
- Participated and presented in the *Research Scholar Conclave 2022* organized by National Institute of Technology Raipur. Title of the paper: **Computational analysis of batch renewal input with correlated batch service discrete-time queueing system**
- Participated in the *27<sup>th</sup> International Conference of International Academy of Physical Sciences (CONIAPS XXVII) on “RECENT ADVANCES IN STATISTICS AND OPERATIONS RESEARCH”*, 2021-- Department of Mathematics, National Institute of Technology Sikkim
- Participated in the *International Webinar Series on “Applications of Stochastic Models and Queueing Theory”*, 2021 -- Department of Mathematics, Central University of Rajasthan
- Participated in the *13<sup>th</sup> International Workshop on Retrial Queues and Related Topics (WRQ 2021@CMS)* – Centre for Research in Mathematics, Department of Mathematics, CMS College of Kottayam
- Participated in the “*Advanced Training in Mathematics Schools*” one month course entitled “*Linear Algebra*”, 2021 -- (National Centre for Mathematics: A joint centre of IIT Bombay and TIFR Mumbai), National Institute of Technology Arunachal Pradesh
- Participated in the *One Week National level “Workshop on Software in Mathematics and Statistics” (WSMS-2021)* -- Department of Mathematics, National Institute of Technology Tiruchirappalli
- Participated in the “*INDIAN WOMEN AND MATHEMATICS (IWM): Regional Workshop on Research and Opportunities*”, 2021, -- Department of Mathematics, Indian Institute of Technology Bhilai
- Participated in the *One week Short Term Course under TEQUIP-III on “Operations Research and its Applications”*, 2020 -- Department of Mathematics, Indian Institute of Technology Delhi
- Participated in the *One week Short Term Course under TEQUIP-III on “Matrix Analysis and its Applications (MAA-2020)”* -- Department of Mathematics, National Institute of Technology Jalandhar
- Participated and presented in the *National Conference on “RECENT TRENDS IN APPLIED AND COMPUTATIONAL MATHEMATICS (RTACM-2019)”* -- Department of Mathematics, National Institute of Technology Raipur and got the *Best Paper Award*. Title of the paper: **Analysis of an infinite buffer discrete-time queueing system with geometric arrival and bulk service facility**
- Participated in the *One Week National Workshop on “Programming, Simulation and Optimization using MATLAB”*, 2019 -- Department of Computer Science and Engineering, National Institute of Technology Raipur.

### Master's Project

**Project Guide:** Prof. Krishnendu Dutta (Principal, Maulana Azad College, Kolkata),  
Prof. Sandip Jana (HOD, Department of Pure Mathematics, Bullygunge Science College,  
University of Calcutta).

**Project title :** Statistical convergence and ideal convergence of real sequence

### Achievements:

- Selected as a Visiting Postdoctoral Fellow at the Computer Systems Group (CSG) in IIIT-Hyderabad, India, in 2024.
- Selected as Assistant Professor in the Alliance School of Applied Mathematics, Faculty of Science & Technology at Alliance University, Bangalore, in 2023.
- **DST INSPIRE (SHE)** scholarship holder from 2011 to 2014
- Award of the West Bengal Govt. Merit-cum-Means Scholarship from Secondary to Master's level
- Awarded certificates of merit at High School for examination performance
- Awarded certificates of academic performance from the government organization

### Linguistic Efficiency:

Speaking as well as writing Bengali, English and Hindi.

**Assurance of my efficiency as well as services at all convenient situations.**



**SIGNATURE**